

Lecture 3

Let $\sim$ be an equivalence relation ^{on} a non-empty set X .

For each $a \in X$, the equivalence class of a is

$$[a] = \{b \in X \mid a \sim b\}.$$

Theorem 1: Let $\sim$ be an equivalence relation, on a non-empty set X . Then the following statements are true:

- i) for each $a \in X$, $a \in [a]$
- ii) for each $a, b \in X$, either $[a] \cap [b] = \emptyset$ or $[a] = [b]$
- iii) $X = \bigcup_{a \in X} [a]$

Proof: i) Since $\sim$ is an equivalence relation, $\sim$ is reflexive.

$\Rightarrow$ For each $a \in X$, $a \sim a$, i.e. $a \in [a]$.

ii) Let $a, b \in X$. If $[a] \cap [b] \neq \emptyset$, we claim $[a] = [b]$.

Suppose $c \in [a] \cap [b]$.

$$c \in [a] \Rightarrow a \sim c$$

$$c \in [b] \Rightarrow b \sim c \Rightarrow c \sim b \quad [\because \sim \text{ is symmetric}]$$

Since $\sim$ is transitive, $a \sim c$ & $c \sim b \Rightarrow a \sim b$.

$$\Rightarrow b \sim a \quad [\because \text{Symmetric}].$$

$$\text{Let } x \in [a] \Rightarrow a \sim x.$$

$$b \sim a \text{ and } a \sim x \Rightarrow b \sim x. \quad [\text{By transitivity}]$$

$$\Rightarrow x \in [b] \Rightarrow [a] \subseteq [b]$$

By similar argument we get $[b] \subseteq [a]$.

$$\Rightarrow [a] = [b].$$

(iii) Since $a \in [a]$ ~~$\forall a \in X$~~ ,

we have $\{a\} \subseteq [a] \subseteq X$.

$$\bigcup_{a \in X} \{a\} \subseteq \bigcup_{a \in X} [a] \subseteq X.$$

$$\Rightarrow X \subseteq \bigcup_{a \in X} [a] \subseteq X.$$

$$\Rightarrow \bigcup_{a \in X} [a] = X.$$

Corollary: An equivalence relation $\sim$ on a non-empty set X , decomposes X into a disjoint union of distinct equivalence classes of X w.r.t. $\sim$.

Proof: We know $X = \bigcup_{a \in X} [a]$.

We also know that equivalence classes are either disjoint or identical. Let $[a_1], [a_2], \dots, [a_t]$ are all the distinct equivalence classes of X w.r.t. $\sim$.

Then

$$X = \bigcup_{i=1}^t [a_i]$$

Hence the claim.

Converse of Theorem 1: Let $\{F_\alpha\}_{\alpha \in I}$ be a partition of X .

Then there exists an equivalence relation on X such that $\{F_\alpha\}_{\alpha \in I}$ is the collection of all distinct equivalence classes wrt. this relation.

Proof: As $\{F_\alpha\}_{\alpha \in I}$ is a partition of X , we have

$$F_\alpha \cap F_\beta = \emptyset \quad \forall \alpha, \beta \in I \text{ s.t. } \alpha \neq \beta.$$

$$\& \bigcup_{\alpha \in I} F_\alpha = X.$$

For $a, b \in X$, define

$a \sim b$ if $a, b \in F_\alpha$ for some $\alpha \in I$.

i) $a \sim a \quad \forall a \in X. \Rightarrow \sim$ is reflexive.

ii) $a \sim b \Rightarrow a, b \in F_\alpha \Rightarrow b \sim a$. So $\sim$ is symmetric.

iii) Let $a \sim b$, $b \sim c$ for some $a, b, c \in X$.

$\Rightarrow a, b \in F_\alpha \quad \& \quad b, c \in F_\beta$ for some $\alpha, \beta \in I$.

$$\Rightarrow b \in F_\alpha \cap F_\beta.$$

$$\Rightarrow \alpha = \beta.$$

$$\Rightarrow a, b, c \in F_\alpha = F_\beta \Rightarrow a \sim c.$$

So $\sim$ is transitive.

Hence $\sim$ is an equivalence relation.

Let $a \in X$, then $\exists \alpha \in I$ s.t. $a \in F_\alpha$.

The equivalence class of a , $[a] = \{b \in X \mid a \sim b\}$
 $= \{b \in X \mid b \in F_\alpha\}$
 $= F_\alpha$.

$\Rightarrow$ If $a \in F_\alpha$ then $[a] = F_\alpha$.

This shows that $\{F_\alpha\}_{\alpha \in I}$ is the collection of all distinct equivalence classes of X w.r.t. $\sim$.

Binary operation

Let X be a non-empty set. A binary operation on X is defined as a mapping

$$\star : X \times X \rightarrow X.$$

$$\text{i.e. } \star(a, b) \in X \quad \forall a, b \in X.$$

We will denote the element $\star(a, b)$ by $a \star b \quad \forall a, b \in X$.

Eg: '+' is a binary operation on $\mathbb{N}$.

$$a + b \in \mathbb{N} \quad \forall a, b \in \mathbb{N}.$$

Also, '.' is a binary operation on $\mathbb{N}$.

Note subtraction is not a binary operation on $\mathbb{N}$

$$\text{as } 2 - 3 = -1 \notin \mathbb{N}.$$

but subtraction is a binary operation on $\mathbb{Z}$.

'-' is not commutative on $\mathbb{Z}$ as

$$2-3 \neq 3-2,$$

'-' is not associative on $\mathbb{Z}$ as

$$(3-2)-1 \neq 3-(2-1)$$

Let $\star$ be a binary operation on a non-empty set X .

$$\text{i.e. } \star : X \times X \rightarrow X.$$

i) An element $e_1 \in X$ is called a left identity of X if $e_1 \star a = a \quad \forall a \in X$.

ii) An element $e_2 \in X$ is called a right identity of X if $a \star e_2 = a \quad \forall a \in X$.

iii) An element $e \in X$ is called an identity of X if $a \star e = a = e \star a \quad \forall a \in X$.

Eg: '+' is a binary operation on $\mathbb{Z}$.

0 is a left identity, right identity and identity of $\mathbb{Z}$.

'.' is a binary operation $\mathbb{N}$. (or $\mathbb{Q} \setminus \{0\}$)

1 is a left identity, right identity and identity of $\mathbb{N}$. (or $\mathbb{Q} \setminus \{0\}$)

Inverse: Let $\star$ be a binary operation on a non-empty set X . ~~An element $a \in X$ is called a left inverse of a if~~

Let e be the identity of X , let $a \in X$.

- i) ~~An element $a_1 \in X$ is called a left inverse of a if~~ An element $a_1 \in X$ is called a left inverse of a if $a_1 \star a = e$.
- ii) An element $a_2 \in X$ is called a right inverse of a if $a \star a_2 = e$.
- iii) An element $a' \in X$ is called an inverse of a if $a \star a' = e = a' \star a$.

And the element a is called invertible.

Ex. Consider $(\mathbb{Z}, +)$.

-2 is the left inverse, right inverse and inverse of 2 .

Consider $(\mathbb{Q} \setminus \{0\}, \cdot)$.

$\frac{1}{3}$ is the left inverse, right inverse and inverse of 3 .

Theorem: Let $\star$ be an associative binary operation, on X .

Let $a \in X$ and e be the identity of X .

i) If a has a left inverse and right inverse, then they are equal.

ii) If $a \in X$ has an inverse, then it is unique.

Proof: i) Let a_1 and a_2 be a left and right inverse of a respectively.

$$\text{i.e. } a_1 \star a = e \quad \text{--- (1)}$$

$$a \star a_2 = e. \quad \text{--- (2)}$$

$$\text{So } a_1 = a_1 \star e = a_1 \star (a \star a_2) \quad [\text{by (2)}]$$

$$= (a_1 \star a) \star a_2 \quad [\text{by associativity}]$$

$$= e \star a_2 \quad [\text{by (1)}]$$

$$= a_2 \quad [\because e \text{ is the identity}]$$

ii) Let a' and a'' be two inverses of a .

$$\text{So } a' \star a = e = a \star a'$$

$$\text{+ } a'' \star a = e = a \star a''$$

So

$$a' = a' \star e \quad [\because e \text{ is the identity}]$$

$$= a' \star (a \star a'') \quad [\because a'' \text{ is an inverse}]$$

$$= (a' \star a) \star a'' \quad [\text{by associativity}]$$

$$= e \star a'' \quad [\because a' \text{ is an inverse}]$$

$$= a''$$

